

Task 4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

Author: Smit Patel

Artificial Intelligence & Machine Learning Internship
@Maincrafts Technology

1. Introduction

In Task 4, the focus transitioned from regression to binary classification, utilizing the Breast Cancer dataset. The core objective was to build robust classification models and evaluate them using advanced metrics rather than relying solely on accuracy, which is often misleading for imbalanced datasets.

2. Evaluation Metric Selection

Accuracy simply measures the percentage of correct predictions. In medical datasets (like cancer prediction), accuracy is insufficient because the cost of a False Negative (missing a malignant tumor) is significantly higher than a False Positive. Therefore, the following metrics were prioritized:

- **Recall (Sensitivity):** Crucial for medical diagnosis. It measures out of all actual positive cases, how many were correctly predicted. High recall ensures fewer false negatives.
- **Precision:** Measures how many of the predicted positive cases were actually positive.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a single metric to balance the trade-off.
- **ROC Curve & AUC:** The Area Under the Curve (AUC) evaluates the model's ability to discriminate between malignant and benign classes across different thresholds.

3. Handling Class Imbalance

Real-world datasets often suffer from class imbalance, where one class heavily outnumbered the other. If ignored, the model tends to predict the majority class, creating an illusion of high accuracy. To address this, the `class_weight='balanced'` parameter was applied to the Logistic Regression model. This automatically adjusts the weights inversely proportional to class frequencies, forcing the model to penalize mistakes on the minority class more heavily.

4. Model Comparison & Final Decision

Two models were trained and compared: Logistic Regression and Decision Tree Classifier.

Logistic Regression provided a highly stable and interpretable baseline. With class weights balanced, it achieved strong recall and a high AUC score, proving highly capable of distinguishing between classes. The Decision Tree Classifier, while capturing non-linear relationships, is generally more prone to overfitting and can be less stable than Logistic Regression for this specific tabular data.

Final Decision: The Balanced Logistic Regression model is selected as the primary classifier due to its stability, interpretability, and robust performance across recall and AUC metrics, making it highly suitable for medical diagnostic predictions.